

Agentic Research & Decision Intelligence Platform

Robust Multi-Agent Research Workflow with Evidence, Algorithms, Formulas, UI Artifacts,
and Reproducible Evaluation

Final Course Project

June 21, 2026

Abstract

This report presents a complete multi-agent research and decision-intelligence platform. The system converts one high-level topic into a reproducible research workflow: planning, evidence retrieval, structured analysis, critique, fact-checking, visualization, report generation, evaluation, presentation export, and persistent memory. The report intentionally includes the mathematical scoring formulation, conditional-routing algorithm, implementation tables, generated architecture figures, UI screenshots, and export artifacts. The default demonstration uses deterministic mock mode so that the project can be graded without paid API keys while remaining extensible to live retrieval and model providers.

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1 Executive Summary

This final report documents an inspectable agentic research platform, not a single chatbot response. The analyzed run used the topic *Multi-agent AI systems for trustworthy clinical decision support*, operated in Deterministic mock/offline mode, retrieved 21 evidence items, generated 14 visual artifacts, and achieved an evaluation score of 93.1/100 with confidence 95.5/100. The main engineering contribution is a transparent workflow in which planning, retrieval, analysis, critique, fact checking, visualization, reporting, evaluation, presentation, and memory are separate stages with explicit state.

2 Research Question and Scope

The project asks whether a graph-based multi-agent system can produce a more trustworthy research artifact than a one-shot prompt. The scope is a university-grade prototype for evidence-grounded research and decision support, demonstrated on trustworthy clinical decision support. It is not a medical device and it does not provide patient-specific recommendations. The deliverable scope includes runnable code, a Streamlit panel, FastAPI endpoints, SQLite memory, generated figures, Markdown/LaTeX/PDF reports, evaluation output, and presentation material.

3 Background and Related Work

Retrieval-augmented generation connects model output to external evidence [14]. ReAct-style agents combine reasoning and tool use [20], while Reflexion and Self-Refine demonstrate iterative critique and revision [12], [1]. Evaluation methods for RAG emphasize faithfulness, relevance, and context use [19]. Clinical decision support research stresses workflow fit, safety, and governance [17], while hallucination surveys show why fluent text alone is not enough [23].

4 System Requirements

The platform is designed around six requirements. First, the default path must be reproducible without paid API keys. Second, each agent decision must be inspectable through typed state. Third, retrieved evidence and claim checks must be visible. Fourth, tables and figures must compile cleanly into LaTeX without overlapping text or distorted images. Fifth, the UI must expose intermediate and final artifacts for live grading. Sixth, scripts and tests must regenerate the report and screenshots from source.

5 System Architecture

The platform has five layers: Streamlit and FastAPI interfaces, graph orchestration, specialized agents, deterministic tools, and persistence/export storage. The figures below are included with constrained width, constrained height, and `keepaspectratio` so page layout remains stable.

Overall System Architecture

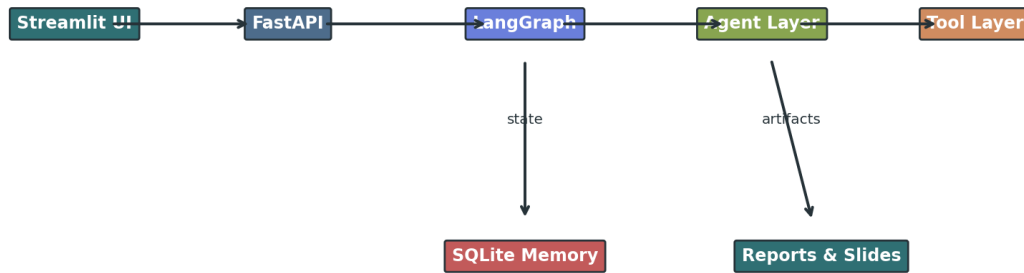


Figure 1: System architecture connecting UI, API, graph orchestration, specialized agents, tools, memory, and exported artifacts.

Conditional Multi-Agent Workflow

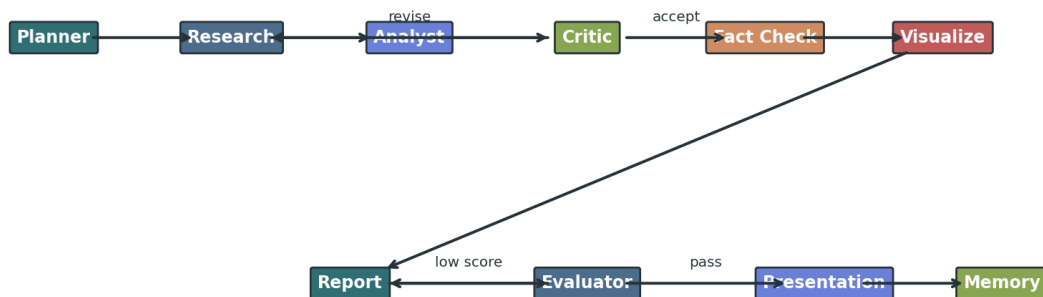


Figure 2: Conditional multi-agent workflow showing planning, retrieval, analysis, critique, fact-checking, report generation, evaluation, presentation, and memory.

Agent Communication Diagram

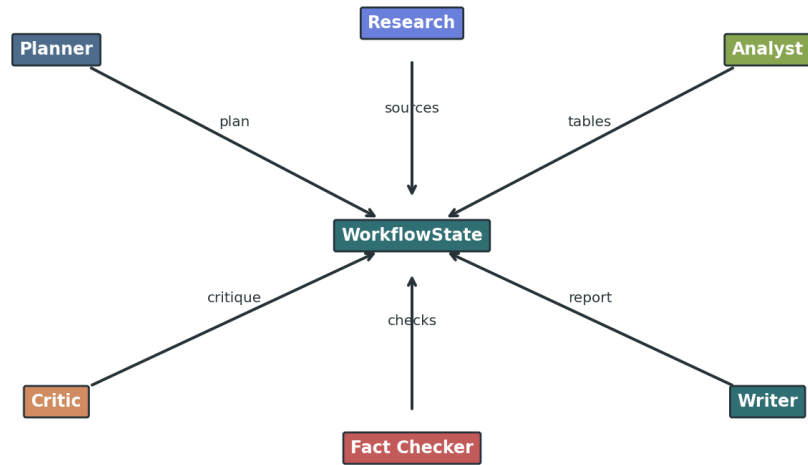


Figure 3: Agent communication through explicit shared WorkflowState fields rather than hidden ad-hoc messages.

LangGraph State Transition Diagram

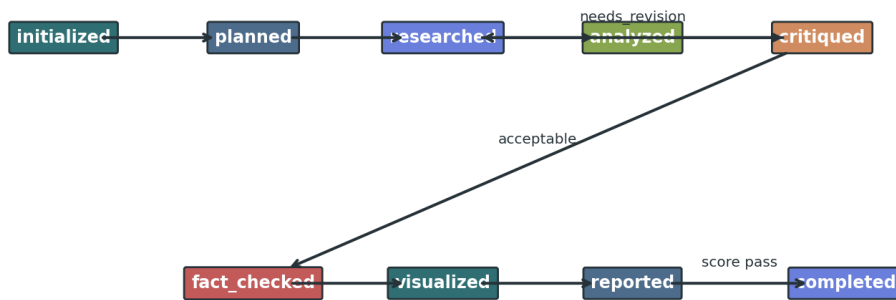


Figure 4: State transition diagram showing how workflow status values move through the graph.

Report Generation Pipeline



Figure 5: Report generation pipeline from evidence and tables to Markdown, LaTeX/PDF, presentation, and evaluation outputs.

Evaluation Pipeline

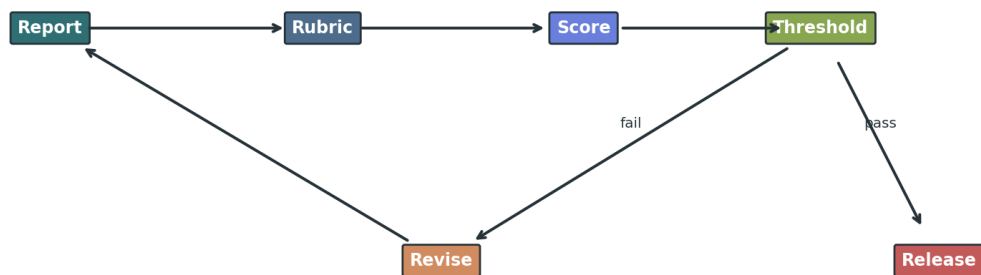


Figure 6: Evaluation pipeline with rubric scoring, thresholding, and optional revision routing.

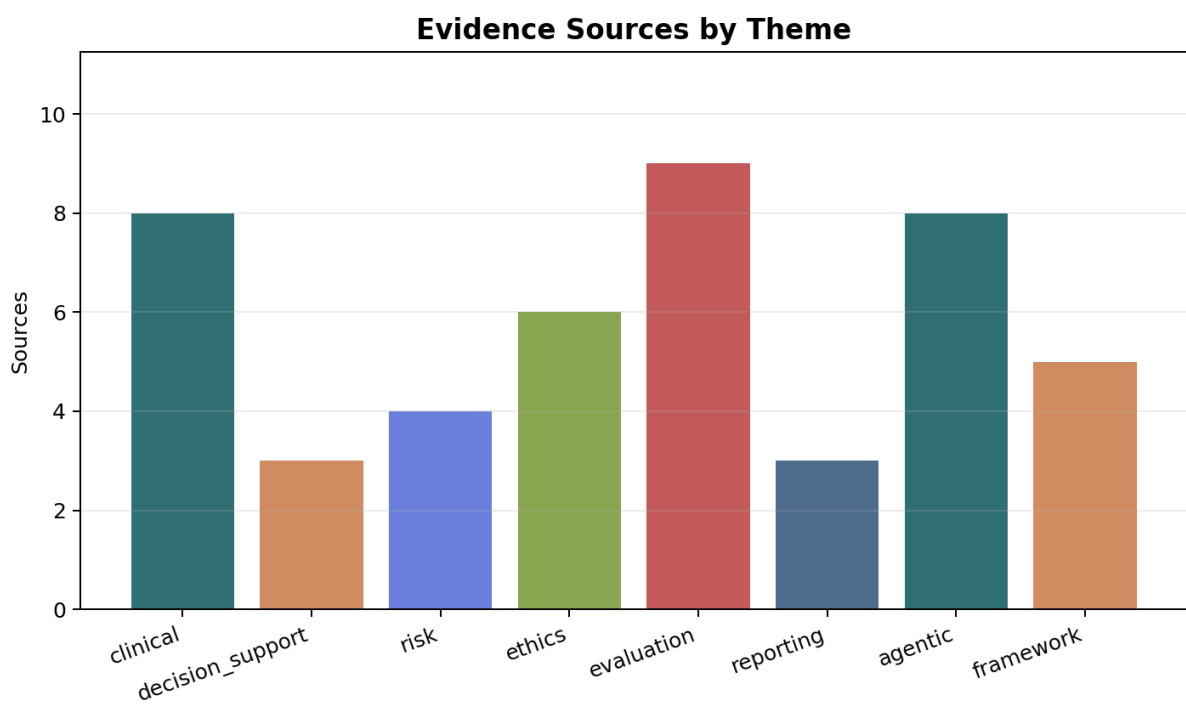


Figure 7: Evidence distribution by theme for the deterministic demonstration corpus.

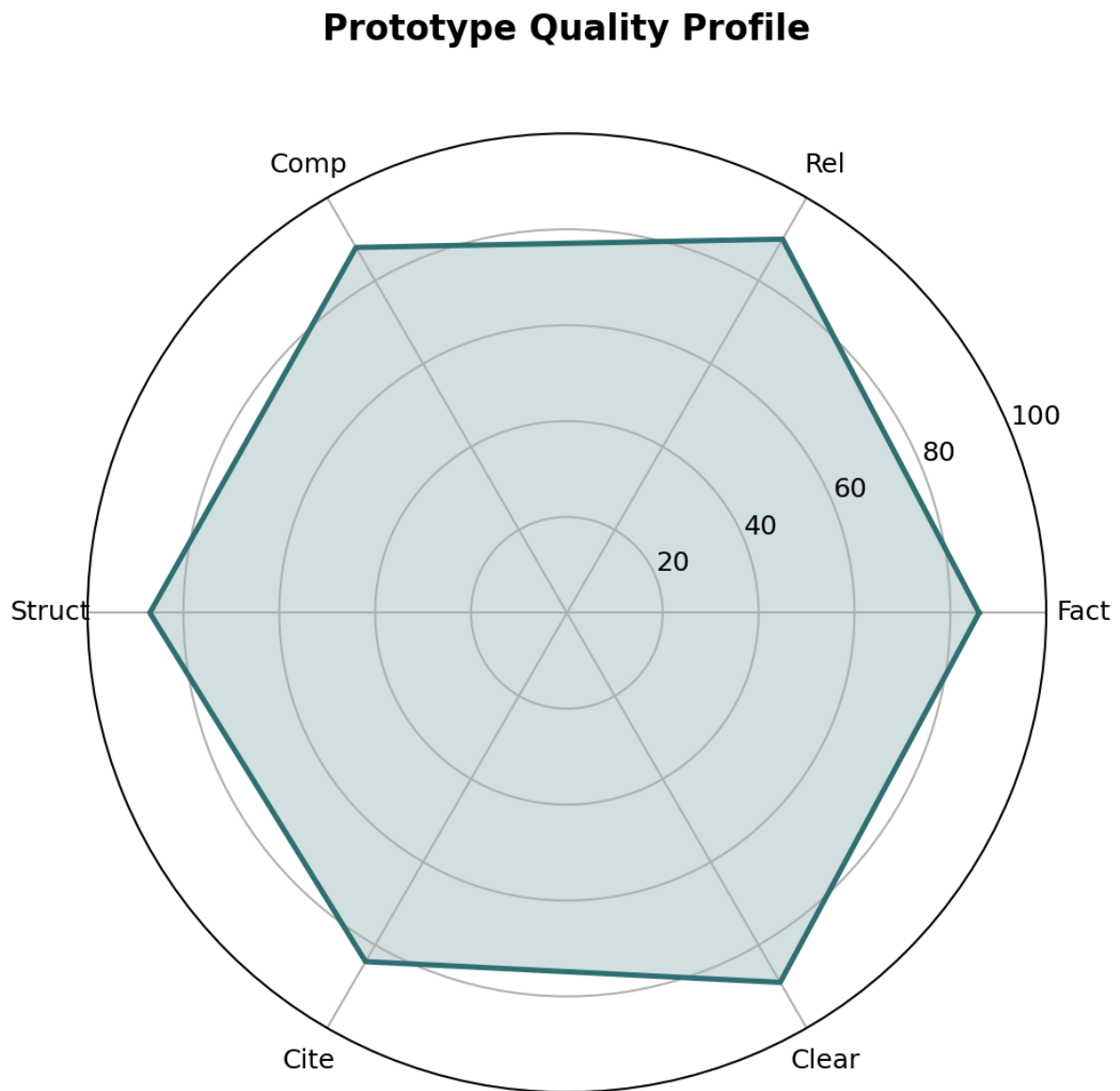


Figure 8: Radar chart of selected quality dimensions for the generated artifact.



Grounded decision intelligence: every recommendation must trace back to evidence, critique, and citation checks.

Figure 9: Conceptual grounding pipeline from user question to evidence-backed synthesis.

6 Agent Responsibilities

Each agent owns one responsibility. Planner decomposes the task; Research retrieves sources; Analyst computes metrics and tables; Critic searches for weaknesses; Fact-Checker grounds claims; Visualization creates charts and diagrams; Report Writer produces Markdown, LaTeX, and PDF artifacts; Evaluator scores output quality; Presentation exports slides; Memory persists the run. The tables below are generated from workflow state and use adaptive column widths to avoid text overlap.

Table 1: Agent Roles

Agent	Responsibility	Input	Output
Planner	Task decomposition and routing	User query	Execution plan (PlanStep list)
Research	Evidence retrieval	Query and plan	Ranked Source objects
Analyst	Metrics, comparisons, formulas	Retrieved sources	Analysis tables and metrics
Critic	Weakness and risk review	Analysis and sources	needs_revision flag and critique
Fact-Checker	Claim grounding	Approved analysis	ClaimCheck objects with citations
Visualization	Figures and diagrams	Workflow state snapshot	PNG, SVG, and Mermaid assets
Report Writer	Academic synthesis	Full workflow state	Markdown, LaTeX, and PDF reports
Evaluator	Rubric scoring	Report and figures	EvaluationScore JSON/Markdown
Presentation	Stakeholder communication	Evaluated state	Markdown and PPTX slides
Memory	Persistence	Completed workflow state	SQLite run record

Table 2: Framework Comparison

Framework	Strengths	Limitations	Suitability for This Project
LangGraph	Typed graph orchestration with conditional edges	Requires explicit state design	Best fit: mirrors critique and evaluation loops
Google ADK	Code-first agent toolkit with Google ecosystem deployment	Younger ecosystem and vendor coupling	Useful for Gemini/Vertex deployments, not required here
AutoGen	Conversational multi-agent patterns	Can become chat-loop centric	Good for research prototypes; less explicit graph control
CrewAI	Role/task abstraction is easy to teach	Less explicit low-level routing	Suitable for sequential crews, not revision gates
Semantic Kernel	Enterprise-friendly plugins and planners	Broad SDK surface area	Strong for Microsoft stacks; heavier than needed

Table 3: API Endpoints

Endpoint	Method	Purpose
/health	GET	Service readiness check
/run	POST	Execute the full agentic workflow
/runs/{run_id}	GET	Retrieve stored run metadata and serialized state
/runs/{run_id}/report	GET	Download the generated Markdown report
/runs/{run_id}/presentation	GET	Download the presentation artifact
/runs/{run_id}/figures	GET	List paths to generated figure files

Table 4: Evaluation Rubric

Criterion	Description	Weight
Factuality	Share of claims supported by retrieved sources	15%
Relevance	Alignment between query, plan, and final report	12%
Completeness	Presence of required sections, tables, figures, and formulas	12%
Structure	Logical organization and academic readability	10%
Citation quality	Traceable references and citation markers	15%
Clarity	Plain-language explanations of technical decisions	10%
Visual quality	Architecture diagrams, charts, and labeled figures	13%
Reproducibility	Mock mode, tests, scripts, Docker, and saved artifacts	13%

Table 5: Limitations and Mitigations

Limitation	Impact	Mitigation
Static mock corpus	Evidence may be stale or incomplete	Deterministic grading; optional live search adapters
Real LLM dependency in production	Non-deterministic outputs without keys	USE MOCK_LLM=true default; provider adapters behind tools
Citation verification limits	Tag-level checks miss entailment errors	Fact-checker gates claims; human review required
Clinical prototype scope	Not validated for patient-specific decisions	Explicit limitations, ethics section, no diagnosis claims
Human review requirement	Automation bias if outputs are trusted blindly	Critic mandates limitations; evaluator threshold gate

7 Workflow and State Model

The workflow is stateful rather than conversationally implicit. **WorkflowState** contains the query, plan, sources, notes, analysis values, tables, critique, revision flags, claim checks, figure paths, report paths, evaluation scores, status, and errors. Conditional routing occurs after the Critic and Evaluator, which makes the graph agentic: weak evidence can route back to Research, and a low report score can route back to Report Writer before finalization.

Table 6: Conditional Routing Rules

Decision point	Condition	Next node
Critic	Evidence is insufficient and revision budget remains	Research Agent
Critic	Evidence is acceptable	Fact-Checker Agent
Evaluator	Quality score $Q < \tau$ where $\tau = 78$	Report Writer Agent
Evaluator	Quality score $Q \geq \tau$	Presentation Agent
Presentation	Slide artifact generated	Memory Agent

8 Mathematical Formulation

The Planner decomposes a user request q into ordered tasks:

$$T(q) = \{t_1, t_2, \dots, t_n\}, \quad t_i = (d_i, a_i, D_i) \quad (1)$$

where d_i is the task description, a_i is the assigned agent, and D_i is the dependency set.

Mock retrieval ranks each source x with a transparent heuristic:

$$S(x, q) = \lambda_1 \text{overlap}(x, q) + \lambda_2 \text{credibility}(x) + \lambda_3 \text{tag_match}(x, q). \quad (2)$$

This keeps grading deterministic while preserving a clear extension point for live search.

The confidence score is modeled as a weighted evidence aggregate:

$$C = 100 \cdot \frac{\sum_{i=1}^n w_i s_i}{\sum_{i=1}^n w_i}, \quad (3)$$

where s_i is a source-quality or coverage signal and w_i is its weight. In this run, $C = 95.5/100$.

The evaluator combines quality dimensions into an overall score:

$$Q = \frac{F + R + M + S + K + L + V + P}{8}, \quad (4)$$

where F is factuality, R relevance, M completeness, S structure, K citation quality, L clarity, V visual quality, and P reproducibility.

The revision gate is:

$$r = \begin{cases} 1, & Q < \tau \text{ and revision budget remains,} \\ 0, & Q \geq \tau \text{ or budget is exhausted,} \end{cases} \quad \tau = 78. \quad (5)$$

When $r = 1$, the graph returns to a previous generation stage instead of silently accepting a weak artifact.

9 Algorithmic Workflow

Listing 1: Algorithm 1: Conditional multi-agent execution

```

Input: query q, threshold tau, maximum revisions m
state <- initialize WorkflowState(q)
state.plan <- Planner(q)
state.sources <- Research(state.plan)
state.analysis <- Analyst(state.sources)
state.critique <- Critic(state.analysis, state.sources)
while state.needs_revision and state.revision_count < m:
    state.revision_count <- state.revision_count + 1
    state.sources <- Research(state.critique)
    state.analysis <- Analyst(state.sources)
    state.critique <- Critic(state.analysis, state.sources)
state.claim_checks <- FactChecker(state.sources, state.analysis)
state.figures <- Visualization(state)
state.report <- ReportWriter(state)
state.evaluation <- Evaluator(state.report, state.figures)
while state.evaluation.total < tau and state.revision_count < m:
    state.revision_count <- state.revision_count + 1
    state.report <- ReportWriter(state)
    state.evaluation <- Evaluator(state.report, state.figures)
state.report <- ReportWriter(state) # final pass with final score
state.presentation <- Presentation(state.report)
Memory.save(state)
return state

```

Listing 2: Algorithm 2: Claim grounding and citation gating

```

Input: candidate claims, retrieved sources
for each claim in candidate claims:
    matching_sources <- sources with related tags, title terms, or summaries
    if matching_sources is empty:
        mark claim unsupported
        attach no generated citation
    else:
        mark claim supported
        attach only retrieved-source citation markers
return claim check list

```

10 Evidence and Citation Strategy

The retrieval component ranks sources by query overlap, credibility, and tags. The fact checker only cites already-retrieved sources and does not invent references. The strongest evidence themes in this run were: evaluation (9), clinical (8), agentic (8), ethics (6), framework (5), risk (4), decision_support (3), reporting (3). Table 7 samples the evidence base used by the generated report.

Table 7: Representative Retrieved Evidence Sources

Type	Year	Credibility	Tags	Title
paper	2020	0.96	clinical, decision_support, risk, ethics	An Overview of Clinical Decision Support Systems: Benefits, Risks, and Strategies for Success
paper	2005	0.94	clinical, decision_support, evaluation	Effects of Computerized Clinical Decision Support Systems on Practitioner Performance and Patient Outcomes
paper	2005	0.94	clinical, decision_support, evaluation	Improving Clinical Practice Using Clinical Decision Support Systems: A Systematic Review of Trials
guideline	2022	0.96	clinical, ethics, reporting, evaluation	DECIDE-AI: New Reporting Guidelines to Bridge the Development-to-Implementation Gap in Clinical Artificial Intelligence
docs	2026	0.84	agentic, framework, adk	Agent Development Kit Multi-Agent Systems Documentation
policy	2021	0.98	clinical, ethics, governance, risk	Ethics and Governance of Artificial Intelligence for Health
guideline	2020	0.97	clinical, ethics, evaluation, reporting	CONSORT-AI Extension: Reporting Guidelines for Clinical Trials of Artificial Intelligence Interventions
guideline	2020	0.97	clinical, ethics, reporting	SPIRIT-AI Extension: Protocol Guidelines for Clinical Trials of Artificial Intelligence Interventions
paper	2023	0.95	clinical, medical, evaluation, llm	Large Language Models Encode Clinical Knowledge
paper	2023	0.91	hallucination, evaluation, risk	Survey of Hallucination in Natural Language Generation
paper	2023	0.84	agentic, multi_agent, framework	AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation
paper	2020	0.94	rag, retrieval, hallucination, evidence	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

11 Implementation Details

The repository separates concerns into `app/agents`, `app/graph`, `app/tools`, `app/models`, `app/storage`, `app/ui`, and `scripts`. The graph runner uses LangGraph when available and a deterministic fallback runner otherwise. Tools handle retrieval, citation formatting, visualization, report generation, presentation export, and filesystem writes. This separation keeps the agents simple and makes the project easier to test.

12 API, Storage, and Artifacts

FastAPI provides health, run, report, presentation, and figure endpoints. SQLite stores run metadata and serialized state for later inspection. Generated artifacts are written to `outputs/` for run-specific files and `report/` for canonical submission files. The canonical report paths are `report/final_report.tex`, `report/final_report.pdf`, `report/final_report.md`, and `report/references.bib`.

13 User Interface and Panel Screenshots

The Streamlit panel exposes the project without requiring graders to read code first. It includes topic input, mock/real mode selection, run execution, workflow inspection, agent outputs, evaluation metrics, figure previews, and download controls. Screenshot capture type: `browser`. Metadata: Captured current Streamlit UI viewport with Playwright/Chromium on port 8511 after seeded mock demo.

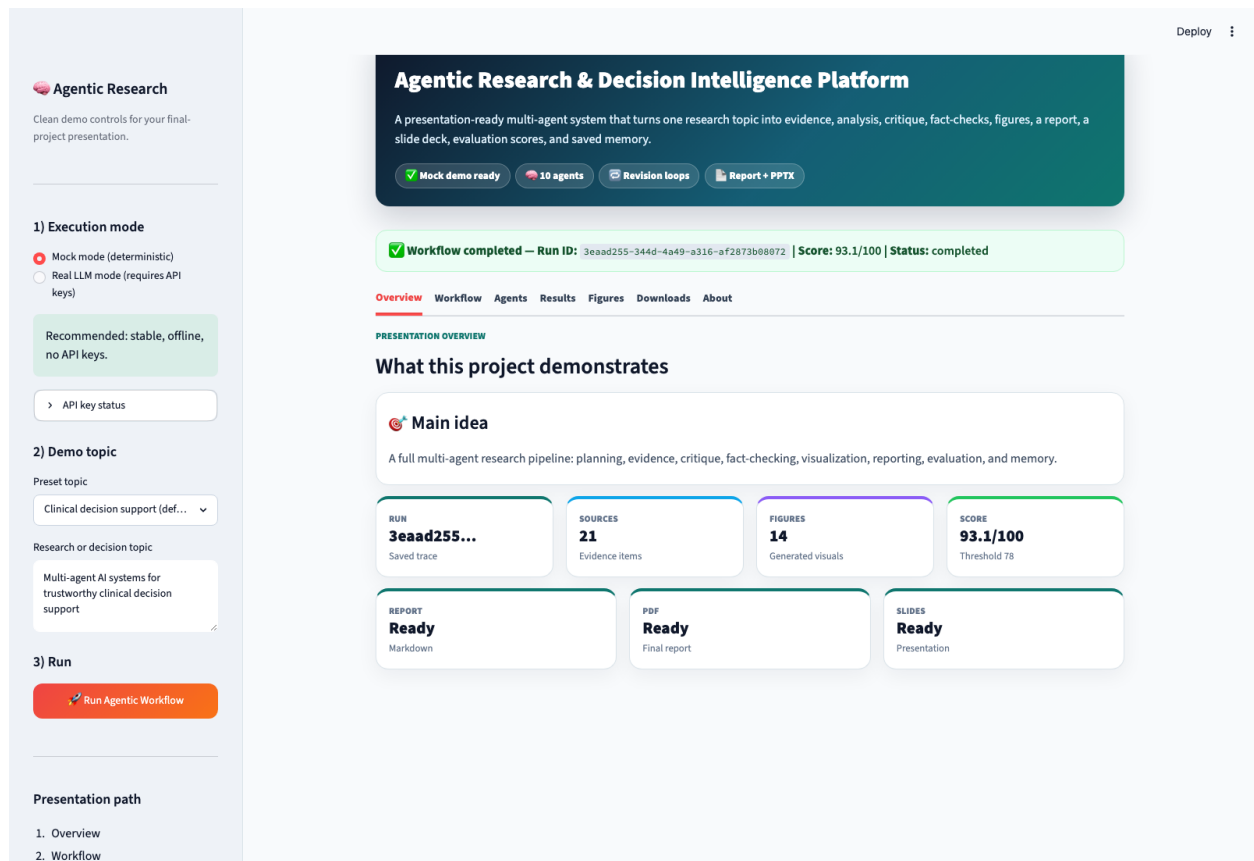


Figure 10: Actual Streamlit Overview tab captured in a headless browser after mock demo run.

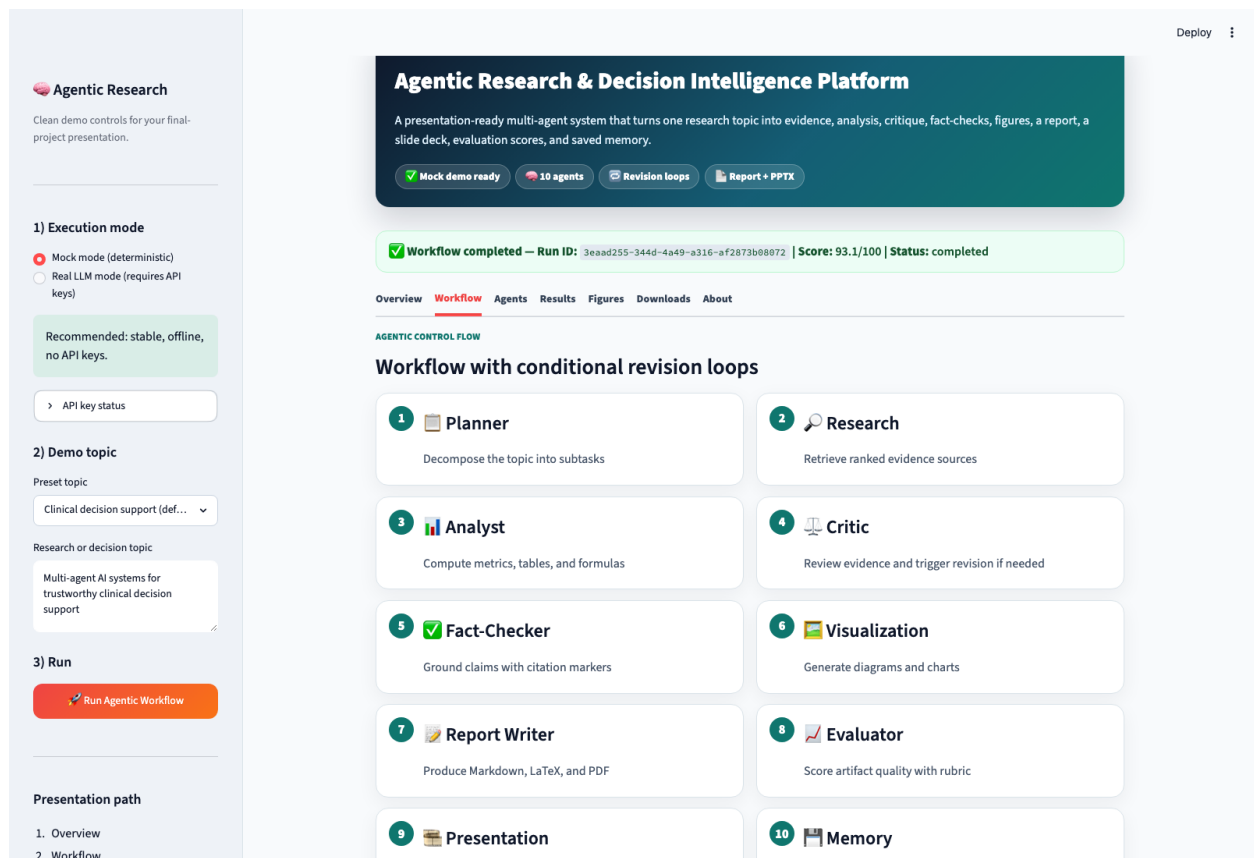


Figure 11: Actual Streamlit Workflow tab captured in a headless browser after mock demo run.

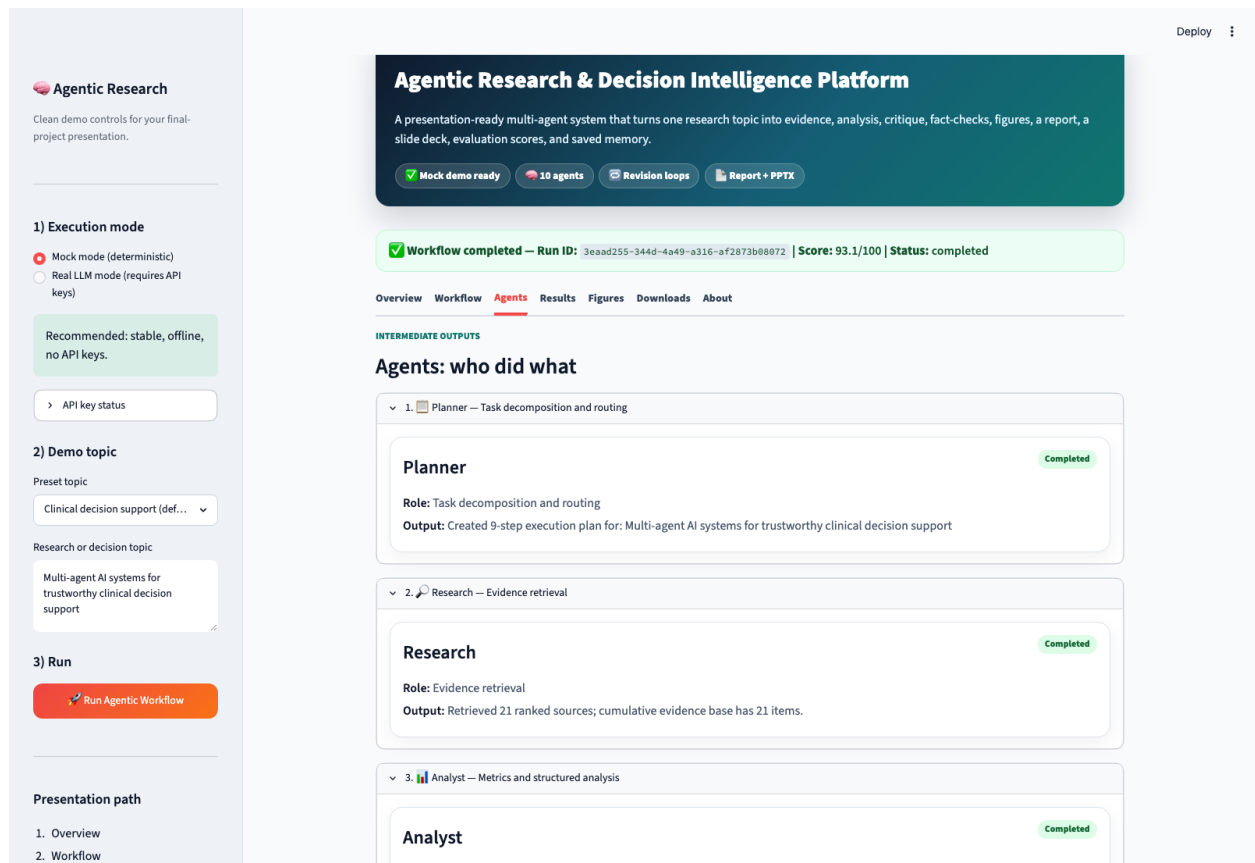


Figure 12: Actual Streamlit Agents tab captured in a headless browser after mock demo run.

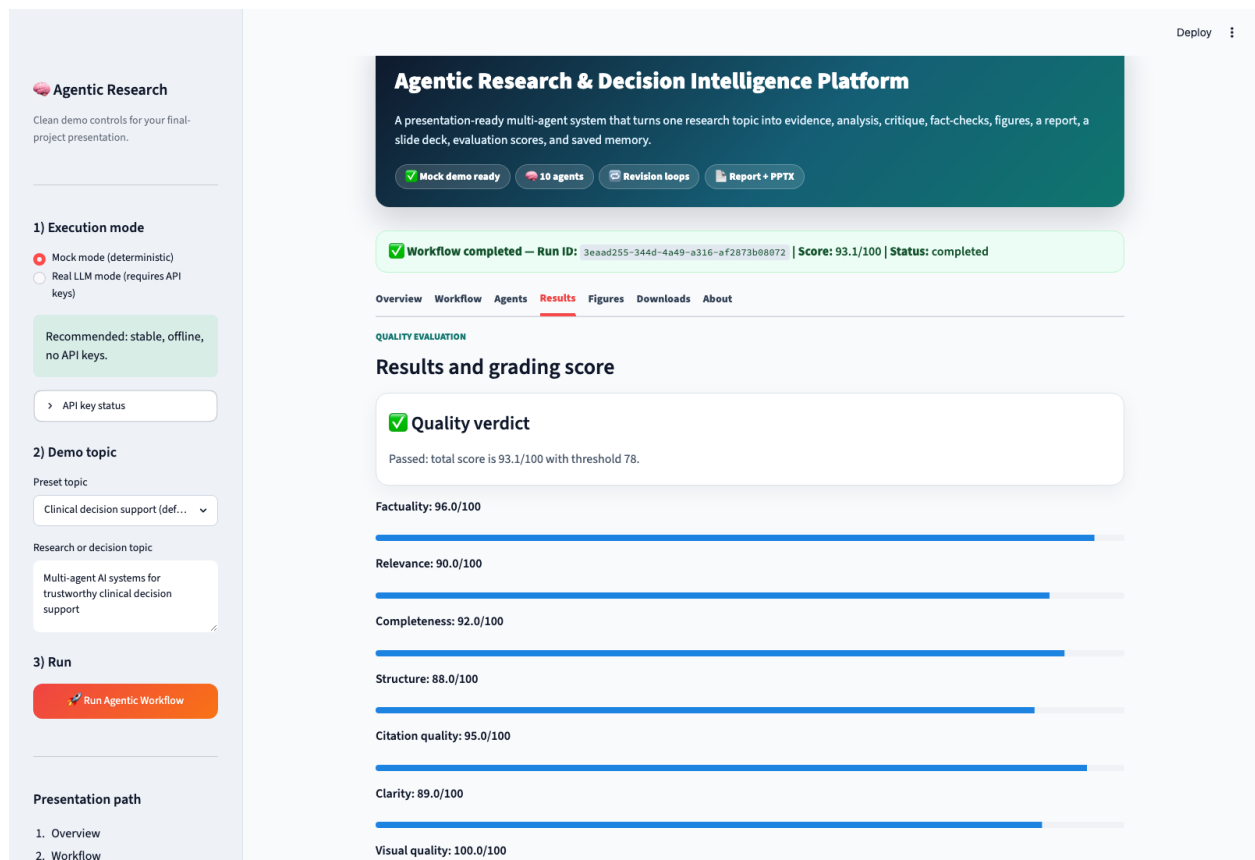


Figure 13: Actual Streamlit Results tab captured in a headless browser after mock demo run.

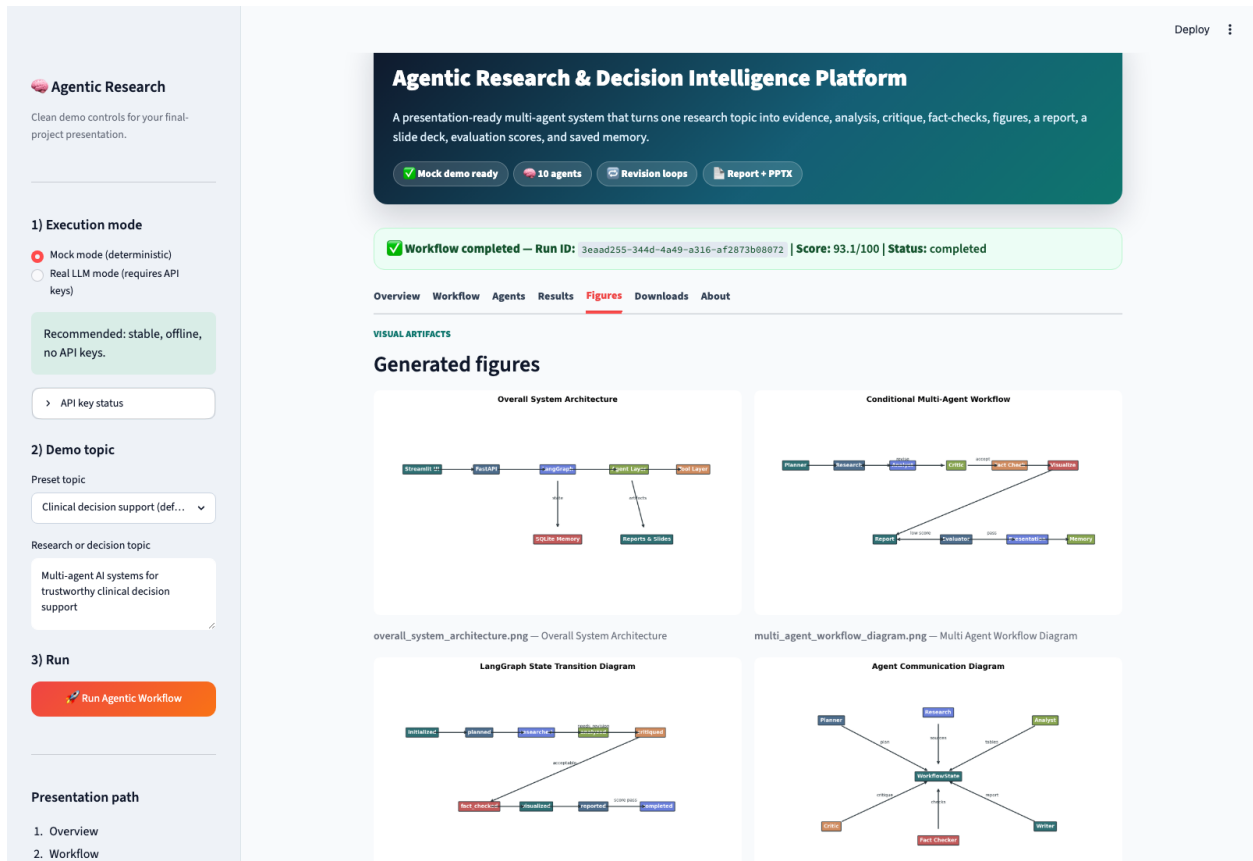


Figure 14: Actual Streamlit Figures tab captured in a headless browser after mock demo run.

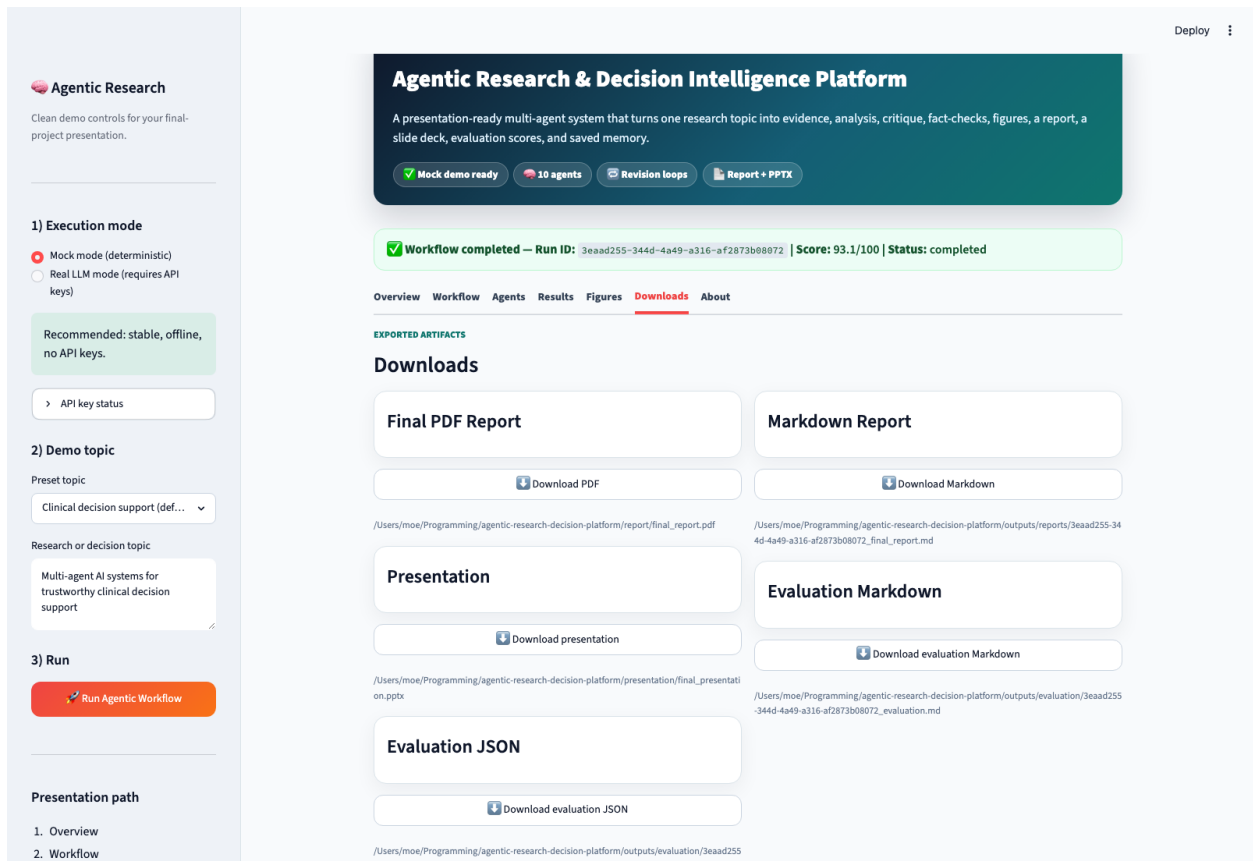


Figure 15: Actual Streamlit Downloads tab captured in a headless browser after mock demo run.

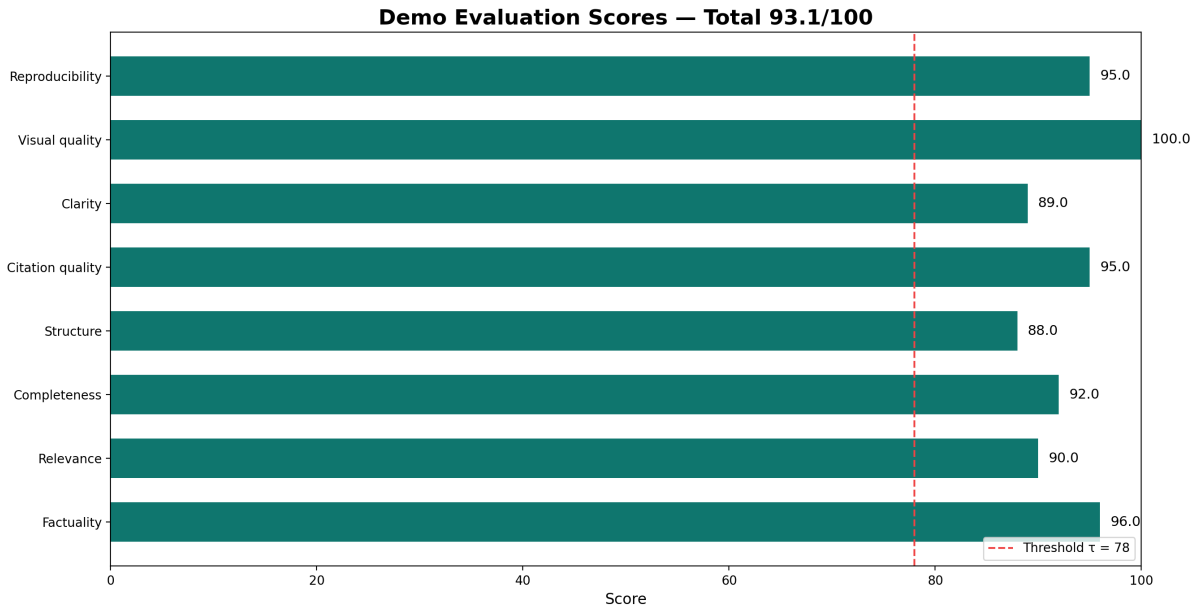


Figure 16: Demo evaluation score panel generated from the reproducible workflow run.

Demo Agent Workflow Result

Run ID: ebd3f228-2cde-4e75-b936-82170141aa03
 Status: completed
 Sources retrieved: 21
 Figures generated: 14
 Evaluation score: 93.1/100

Workflow pipeline:

1. Planner — Decompose the topic into subtasks
2. Research — Retrieve ranked evidence sources
3. Analyst — Compute metrics, tables, and formulas
4. Critic — Review evidence and trigger revision if needed
5. Fact-Checker — Ground claims with citation markers
6. Visualization — Generate diagrams and charts
7. Report Writer — Produce Markdown, LaTeX, and PDF
8. Evaluator — Score artifact quality with rubric
9. Presentation — Generate slide deck
10. Memory — Persist run metadata to SQLite

Figure 17: Demo workflow result panel summarizing run ID, status, sources, figures, and agent sequence.

Demo Generated Figures

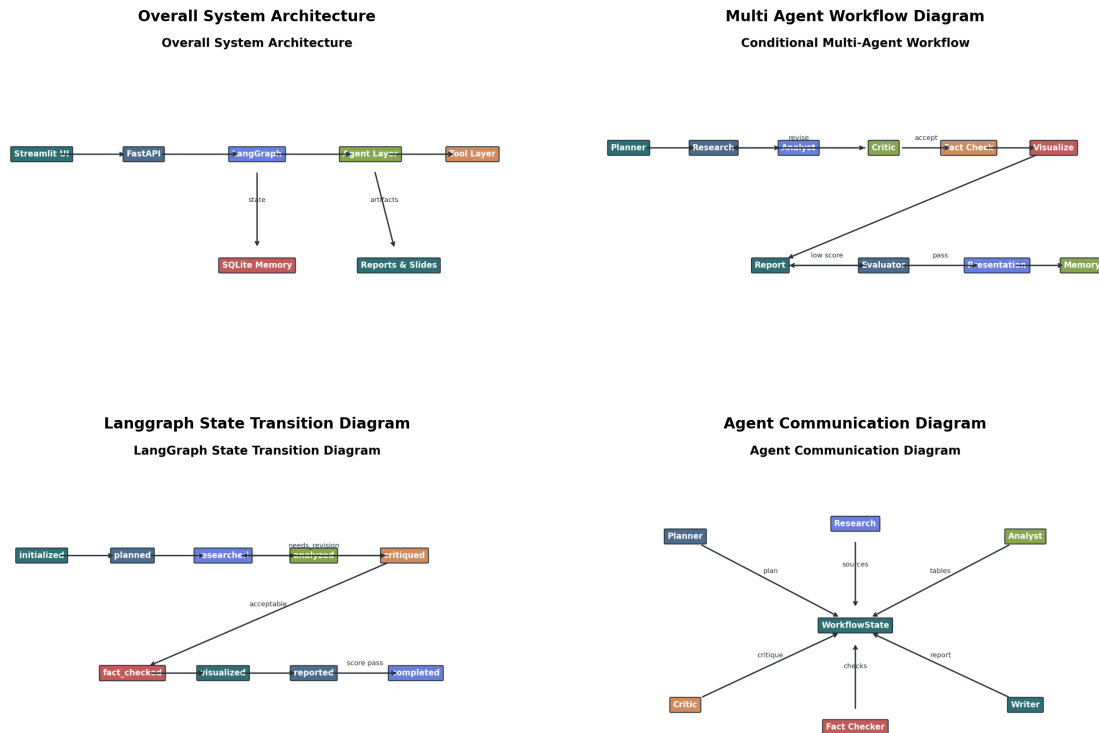


Figure 18: Demo gallery of generated figures used by the final report.

Demo Report Output Preview

Agentic Research & Decision Intelligence Platform
Final Course Project Report

Topic: Multi-agent AI systems for trustworthy clinical decision support
Run ID: ebd3f228-2cde-4e75-b936-82170141aa03
Quality score: 93.1/100
Sources: 21 | Figures: 14

Abstract

This report documents a complete multi-agent research and decision-intelligence platform. The system converts a broad topic into a reproducible workflow: planning, evidence retrieval, analysis, critique, fact-checking, visualization, report generation, evaluation, presentation export, and persistent memory. The revised report emphasizes mathematical formulation, conditional algorithms, readable tables, correctly scaled figures, panel screenshots, and reproducible demo commands.

Included sections

1. Problem Definition
2. System Overview
3. Agentic Architecture
4. Mathematical Formulation
5. Algorithmic Workflow
6. Evidence and Retrieval
7. Tables
8. Figures and Images
9. UI Panel Screenshots
10. Evaluation
- ... and 6 more sections

Figure 19: Demo report preview showing generated academic sections and run metadata.

14 Report Generation and LaTeX Formatting

The LaTeX generator now uses `tabularx`, an adaptive Y column type, `adjustbox`, smaller table padding, increased row spacing, and constrained figure dimensions. Inline math is preserved while regular text is escaped, so table cells can contain formulas such as $Q < \tau$ without being converted into broken text. Figures and screenshots use `keepaspectratio` with a maximum height, which prevents stretched diagrams and tiny unreadable page images.

15 Evaluation Methodology

The evaluator scores factuality, relevance, completeness, structure, citation quality, clarity, visual quality, and reproducibility. These dimensions are important because a research assistant should be graded on traceability and artifact quality, not just prose fluency. The final score in this run is 93.1/100.

Table 8: Final Evaluation Scores

Dimension	Score	Interpretation
Factuality	96.0	Groundedness and unsupported-claim control
Relevance	90.0	Alignment with the selected topic
Completeness	92.0	Coverage of required deliverables
Structure	88.0	Organization and readability
Citation quality	95.0	Evidence traceability
Clarity	89.0	Presentation quality
Visual quality	100.0	Figure and UI artifact quality
Reproducibility	95.0	Repeatable mock-mode execution
Total	93.1	Overall artifact score

16 Results

The mock run completed with 21 sources, 14 figure records, a canonical LaTeX source, a compiled PDF, demo screenshots, result images, evaluation JSON/Markdown, and presentation output. The report is intentionally artifact-heavy: tables document agent behavior and API design; formulas document scoring; algorithms document control flow; figures document architecture and UI behavior.

17 Claim Checks

- **Supported:** Retrieval grounding can reduce unsupported generation by forcing reports to cite external evidence. ((Patrick Lewis, 2020), (Shahul Es, 2023)). Supported by tagged source evidence.
- **Supported:** Clinical decision support requires human oversight and careful workflow integration. ((Amit X. Garg, 2005), (Kensaku Kawamoto, 2005), (Reed T. Sutton, 2020)). Supported by tagged source evidence.
- **Supported:** Conditional multi-agent graphs make critique and revision loops explicit. ((Google, 2026), (Qingyun Wu, 2023), (Shunyu Yao, 2022)). Supported by tagged source evidence.
- **Supported:** Evaluation should include factuality, relevance, structure, citation quality, and reproducibility. ((Amit X. Garg, 2005), (Brennan Vasey, 2022), (Kensaku Kawamoto, 2005)). Supported by tagged source evidence.

- **Supported:** The prototype is not a medical device and should not produce patient-specific diagnosis. ((Brennan Vasey, 2022), (Reed T. Sutton, 2020), (World Health Organization, 2021)). Supported by tagged source evidence.

18 Critic Notes

- Evidence coverage is adequate for a prototype, but deployment claims must remain bounded.
- The system should emphasize clinician oversight, auditability, and uncertainty rather than autonomous diagnosis.
- The mock corpus is suitable for reproducible testing but must be refreshed for real clinical use.

19 Error Analysis

Expected errors include stale sources, unsupported claims, shallow critique, malformed citations, and broken rendering. The system mitigates these through deterministic retrieval, explicit critic notes, citation gating, evaluation thresholds, generated screenshots, and a render-and-check PDF workflow. For production, passage-level entailment and source freshness checks should be added before any high-stakes deployment.

20 Ethical and Safety Considerations

The clinical scenario is used only as a demanding research context. The system must not be used for diagnosis or treatment. Real deployment would require privacy controls, audit logs, security review, institution-approved source corpora, clinician accountability, bias evaluation, and human approval gates. The report therefore frames clinical decision support as a governance and workflow problem, not merely a model-quality problem.

21 Testing and Reproducibility

The project can be regenerated with the commands below. Mock mode is the grading default because it produces repeatable evidence, figures, screenshots, evaluation files, and reports.

Listing 3: Local verification commands

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
export USE MOCK_LLM=true
python scripts/capture_ui_screenshots.py
python scripts/generate_latex_report.py
python scripts/build_report_pdf.py
pytest
```

22 Limitations

The current corpus is static, the fact checker is metadata-level rather than entailment-level, and the evaluator uses deterministic heuristics. The UI is intended for demonstration rather than production operations. These limitations are acceptable for a final project because they make grading reproducible, but they must be addressed before real deployment.

23 Future Work

Future work should add live retrieval adapters, vector search over approved corpora, stronger natural-language entailment, asynchronous job queues, authentication, richer audit trails, human review checkpoints, and benchmarking against clinical reporting standards. The architecture already isolates these improvements behind agent and tool contracts.

24 Conclusion

The platform demonstrates a complete, reproducible, and inspectable agentic research workflow. The updated LaTeX generator produces a fuller academic report with robust tables, preserved mathematical notation, explicit algorithms, constrained figures, refreshed UI screenshots, and clearer explanations of how the multi-agent system works.

References

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